

## HW3

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|                        | C=1   | C=5   | C=10  | C=50  | C=100 | C=500 | C=1000 |
|------------------------|-------|-------|-------|-------|-------|-------|--------|
| 1.05 <sup>(-100)</sup> | 61.33 | 61.33 | 61.33 | 61.33 | 61.33 | 61.33 | 61.33  |
| 1.05 <sup>(-95)</sup>  | 80.67 | 80.67 | 80.67 | 80.67 | 80.67 | 80.67 | 80.67  |
| 1.05 <sup>(-90)</sup>  | 64    | 64    | 64    | 64    | 64    | 64    | 64     |
| 1.05 <sup>(-85)</sup>  | 66.67 | 66.67 | 66.67 | 66.67 | 66.67 | 66.67 | 66.67  |
| 1.05 <sup>(-80)</sup>  | 57.33 | 57.33 | 57.33 | 57.33 | 57.33 | 57.33 | 57.33  |
| 1.05 <sup>(-75)</sup>  | 57.33 | 52    | 52    | 52    | 52    | 52    | 52     |
| 1.05 <sup>(-70)</sup>  | 47.33 | 47.33 | 47.33 | 47.33 | 47.33 | 47.33 | 47.33  |
| 1.05 <sup>(-65)</sup>  | 50    | 54.67 | 54.67 | 54.67 | 54.67 | 54.67 | 54.67  |
| 1.05 <sup>(-60)</sup>  | 57.33 | 57.33 | 57.33 | 57.33 | 57.33 | 57.33 | 57.33  |
| 1.05 <sup>(-55)</sup>  | 64    | 64    | 64    | 64    | 64    | 64    | 64     |
| 1.05 <sup>(-50)</sup>  | 65.33 | 66.67 | 66.67 | 66.67 | 66.67 | 66.67 | 66.67  |
| 1.05 <sup>(-45)</sup>  | 74    | 75.33 | 75.33 | 75.33 | 75.33 | 75.33 | 75.33  |
| 1.05 <sup>(-40)</sup>  | 84.67 | 85.33 | 85.33 | 85.33 | 85.33 | 85.33 | 85.33  |
| 1.05 <sup>(-35)</sup>  | 90.67 | 90    | 90    | 90    | 90    | 90    | 90     |
| 1.05 <sup>(-30)</sup>  | 94.67 | 93.33 | 93.33 | 93.33 | 93.33 | 93.33 | 93.33  |
| 1.05 <sup>(-25)</sup>  | 95.33 | 95.33 | 95.33 | 95.33 | 95.33 | 95.33 | 95.33  |
| 1.05 <sup>(-20)</sup>  | 95.33 | 94.67 | 95.33 | 95.33 | 95.33 | 95.33 | 95.33  |
| 1.05 <sup>(-15)</sup>  | 95.33 | 95.33 | 95.33 | 96.67 | 96.67 | 96.67 | 96.67  |
| 1.05 <sup>(-10)</sup>  | 94.67 | 95.33 | 95.33 | 96    | 96.67 | 96.67 | 96.67  |
| 1.05 <sup>(-5)</sup>   | 96    | 95.33 | 95.33 | 94.67 | 94.67 | 95.33 | 95.33  |
| 1.05 <sup>(0)</sup>    | 95.33 | 95.33 | 94.67 | 95.33 | 95.33 | 96    | 96     |
| 1.05 <sup>(5)</sup>    | 96    | 95.33 | 96.67 | 96    | 96.67 | 95.33 | 96     |
| 1.05 <sup>(10)</sup>   | 95.33 | 96    | 96    | 94.67 | 96    | 95.33 | 94.67  |
| 1.05 <sup>(15)</sup>   | 95.33 | 96    | 95.33 | 94    | 95.33 | 95.33 | 94.67  |
| 1.05 <sup>(20)</sup>   | 95.33 | 96    | 96    | 97.33 | 94.67 | 95.33 | 95.33  |
| 1.05 <sup>(25)</sup>   | 94.67 | 95.33 | 96    | 98    | 97.33 | 95.33 | 95.33  |
| 1.05 <sup>(30)</sup>   | 94.67 | 96.67 | 95.33 | 96    | 98    | 96    | 94.67  |
| 1.05 <sup>(35)</sup>   | 93.33 | 95.33 | 98    | 96.67 | 98.67 | 96.67 | 96     |
| 1.05 <sup>(40)</sup>   | 92    | 95.33 | 96    | 96.67 | 96.67 | 98    | 96     |
| 1.05 <sup>(45)</sup>   | nan   | 94.67 | 95.33 | 96    | 96.67 | 98.67 | 97.33  |
| 1.05 <sup>(50)</sup>   | nan   | 93.33 | 94.67 | 95.33 | 97.33 | 98.67 | 98     |
| 1.05 <sup>(55)</sup>   | nan   | 90.67 | 93.33 | 96.67 | 96    | 96.67 | 98.67  |



C 的決策邊界窄，則對於分類錯誤有較低的容忍度，此時對於 outlier 敏感，可能會過度擬合。這兩者都會降低在測試資料上的分類率。

分析 sigma 在適當範圍( $1.05^{20} \sim 1.05^{50}$ )，可以發現此範圍內，CR 以 C 由小到大為低、高、低，代表 C 較小時無法精確正確分類資料，C 大時對於雜訊或離群值敏感，皆會降低分類率。其他的範圍則可發現分類率會隨 C 的大小上升或下降。

2. sigma 的大小的改變與分類率是否有關係？若有，請探討 sigma 的差異與特徵的數值有什麼關聯性？

Sigma 的變化代表訓練資料透過不同的映射函數映射到高維空間，導致資料在高維空間的分布不同。由 grid search 的數據可以應證，在相同的 C 下，sigma 對於分類率有相當大的影響。

1. sigma 較小時， $K(x_i, x_j) \approx 0$ ，代表在映射後每個資料點的影響半徑非常窄，訓練資料密集，模型只關注局部特徵，導致嚴重的過度擬合，使分類率很低。
  2. sigma 較大時， $K(x_i, x_j)$ 能算出有意義的數值，映射後的資料分布較廣泛，並能找到最佳解的位置。但過大的話則  $K(x_i, x_j) \approx 1$ ，訓練資料則過於發散，而失去鑑別度，導致 underfitting，分類率下降，甚至無法求出 bias。
3. 若分析過程不採用 two-fold cross validation，則分類率是否會更高？請探討之。

下圖分別代表 1st-fold 跟 2nd-fold 的分類率。

若不採用 two-fold CV，分類率在某些組合會看起來更高，如 1st-fold 在  $1.05^{(-100)} \sim 1.05^{(-30)}$  的 CR 大於 2nd-fold，而 2nd-fold 在  $1.05^{(-31)} \sim 1.05^{(100)}$  的 CR 幾乎皆大於 1st-fold。每一次 fold 使用的訓練與測試資料不同，因此模型在某些參數範圍內更適合某一 fold 的資料分布，而在另一個 fold 中則表現較差。

這也代表如果只使用單一個 fold 來評估 CR，會比較容易得到樂觀的結果，但 CV 會將兩個 fold 的優劣平均，使所選的超參數更具泛化性。

|            | C=1   | C=5   | C=10  | C=50  | C=100 | C=500 | C=1000 |            | C=1   | C=5   | C=10  | C=50  | C=100  | C=500 | C=1000 |
|------------|-------|-------|-------|-------|-------|-------|--------|------------|-------|-------|-------|-------|--------|-------|--------|
| σ=0.0076   | 62.67 | 62.67 | 62.67 | 62.67 | 62.67 | 62.67 | 62.67  | σ=0.0076   | 60.00 | 60.00 | 60.00 | 60.00 | 60.00  | 60.00 | 60.00  |
| σ=0.0097   | 82.67 | 82.67 | 82.67 | 82.67 | 82.67 | 82.67 | 82.67  | σ=0.0097   | 78.67 | 78.67 | 78.67 | 78.67 | 78.67  | 78.67 | 78.67  |
| σ=0.0124   | 61.33 | 61.33 | 61.33 | 61.33 | 61.33 | 61.33 | 61.33  | σ=0.0124   | 66.67 | 66.67 | 66.67 | 66.67 | 66.67  | 66.67 | 66.67  |
| σ=0.0158   | 65.33 | 65.33 | 65.33 | 65.33 | 65.33 | 65.33 | 65.33  | σ=0.0158   | 68.00 | 68.00 | 68.00 | 68.00 | 68.00  | 68.00 | 68.00  |
| σ=0.0202   | 65.33 | 65.33 | 65.33 | 65.33 | 65.33 | 65.33 | 65.33  | σ=0.0202   | 49.33 | 49.33 | 49.33 | 49.33 | 49.33  | 49.33 | 49.33  |
| σ=0.0258   | 65.33 | 65.33 | 65.33 | 65.33 | 65.33 | 65.33 | 65.33  | σ=0.0258   | 49.33 | 38.67 | 38.67 | 38.67 | 38.67  | 38.67 | 38.67  |
| σ=0.0329   | 45.33 | 45.33 | 45.33 | 45.33 | 45.33 | 45.33 | 45.33  | σ=0.0329   | 49.33 | 49.33 | 49.33 | 49.33 | 49.33  | 49.33 | 49.33  |
| σ=0.0419   | 49.33 | 58.67 | 58.67 | 58.67 | 58.67 | 58.67 | 58.67  | σ=0.0419   | 50.67 | 50.67 | 50.67 | 50.67 | 50.67  | 50.67 | 50.67  |
| σ=0.0535   | 60.00 | 60.00 | 60.00 | 60.00 | 60.00 | 60.00 | 60.00  | σ=0.0535   | 54.67 | 54.67 | 54.67 | 54.67 | 54.67  | 54.67 | 54.67  |
| σ=0.0683   | 68.00 | 68.00 | 68.00 | 68.00 | 68.00 | 68.00 | 68.00  | σ=0.0683   | 60.00 | 60.00 | 60.00 | 60.00 | 60.00  | 60.00 | 60.00  |
| σ=0.0872   | 70.67 | 72.00 | 72.00 | 72.00 | 72.00 | 72.00 | 72.00  | σ=0.0872   | 60.00 | 61.33 | 61.33 | 61.33 | 61.33  | 61.33 | 61.33  |
| σ=0.1113   | 78.67 | 80.00 | 80.00 | 80.00 | 80.00 | 80.00 | 80.00  | σ=0.1113   | 69.33 | 70.67 | 70.67 | 70.67 | 70.67  | 70.67 | 70.67  |
| σ=0.142    | 88.00 | 89.33 | 89.33 | 89.33 | 89.33 | 89.33 | 89.33  | σ=0.142    | 81.33 | 81.33 | 81.33 | 81.33 | 81.33  | 81.33 | 81.33  |
| σ=0.1813   | 92.00 | 90.67 | 90.67 | 90.67 | 90.67 | 90.67 | 90.67  | σ=0.1813   | 89.33 | 89.33 | 89.33 | 89.33 | 89.33  | 89.33 | 89.33  |
| σ=0.2314   | 94.67 | 93.33 | 93.33 | 93.33 | 93.33 | 93.33 | 93.33  | σ=0.2314   | 94.67 | 93.33 | 93.33 | 93.33 | 93.33  | 93.33 | 93.33  |
| σ=0.2953   | 93.33 | 94.67 | 94.67 | 94.67 | 94.67 | 94.67 | 94.67  | σ=0.2953   | 97.33 | 96.00 | 96.00 | 96.00 | 96.00  | 96.00 | 96.00  |
| σ=0.3769   | 94.67 | 94.67 | 94.67 | 94.67 | 94.67 | 94.67 | 94.67  | σ=0.3769   | 96.00 | 94.67 | 96.00 | 96.00 | 96.00  | 96.00 | 96.00  |
| σ=0.481    | 94.67 | 96.00 | 96.00 | 96.00 | 96.00 | 96.00 | 96.00  | σ=0.481    | 96.00 | 94.67 | 94.67 | 97.33 | 97.33  | 97.33 | 97.33  |
| σ=0.6139   | 93.33 | 94.67 | 96.00 | 96.00 | 96.00 | 96.00 | 96.00  | σ=0.6139   | 96.00 | 96.00 | 94.67 | 96.00 | 97.33  | 97.33 | 97.33  |
| σ=0.7835   | 96.00 | 94.67 | 94.67 | 94.67 | 94.67 | 94.67 | 94.67  | σ=0.7835   | 96.00 | 96.00 | 96.00 | 94.67 | 94.67  | 96.00 | 96.00  |
| σ=1.0      | 94.67 | 94.67 | 93.33 | 96.00 | 96.00 | 96.00 | 96.00  | σ=1.0      | 96.00 | 96.00 | 96.00 | 94.67 | 94.67  | 96.00 | 96.00  |
| σ=1.2763   | 94.67 | 93.33 | 97.33 | 96.00 | 96.00 | 96.00 | 96.00  | σ=1.2763   | 97.33 | 97.33 | 96.00 | 96.00 | 97.33  | 94.67 | 96.00  |
| σ=1.6289   | 93.33 | 94.67 | 94.67 | 93.33 | 96.00 | 94.67 | 94.67  | σ=1.6289   | 97.33 | 97.33 | 97.33 | 96.00 | 96.00  | 96.00 | 94.67  |
| σ=2.0789   | 93.33 | 94.67 | 93.33 | 92.00 | 94.67 | 93.33 | 93.33  | σ=2.0789   | 97.33 | 97.33 | 97.33 | 96.00 | 96.00  | 97.33 | 96.00  |
| σ=2.6533   | 94.67 | 94.67 | 94.67 | 98.67 | 93.33 | 93.33 | 93.33  | σ=2.6533   | 96.00 | 97.33 | 97.33 | 96.00 | 96.00  | 97.33 | 97.33  |
| σ=3.3864   | 93.33 | 94.67 | 94.67 | 98.67 | 98.67 | 94.67 | 93.33  | σ=3.3864   | 96.00 | 96.00 | 97.33 | 97.33 | 96.00  | 96.00 | 97.33  |
| σ=4.3219   | 94.67 | 96.00 | 94.67 | 94.67 | 98.67 | 94.67 | 93.33  | σ=4.3219   | 94.67 | 97.33 | 96.00 | 97.33 | 97.33  | 97.33 | 96.00  |
| σ=5.516    | 92.00 | 94.67 | 98.67 | 94.67 | 98.67 | 96.00 | 94.67  | σ=5.516    | 94.67 | 96.00 | 97.33 | 98.67 | 98.67  | 97.33 | 97.33  |
| σ=7.04     | 90.67 | 94.67 | 94.67 | 94.67 | 94.67 | 98.67 | 94.67  | σ=7.04     | 93.33 | 96.00 | 97.33 | 98.67 | 98.67  | 97.33 | 97.33  |
| σ=8.985    | NaN   | 94.67 | 96.00 | 94.67 | 94.67 | 98.67 | 97.33  | σ=8.985    | NaN   | 94.67 | 94.67 | 97.33 | 98.67  | 98.67 | 97.33  |
| σ=11.4674  | NaN   | 93.33 | 93.33 | 94.67 | 94.67 | 98.67 | 98.67  | σ=11.4674  | NaN   | 93.33 | 96.00 | 96.00 | 100.00 | 98.67 | 97.33  |
| σ=14.6356  | NaN   | 90.67 | 93.33 | 96.00 | 94.67 | 94.67 | 98.67  | σ=14.6356  | NaN   | 90.67 | 93.33 | 97.33 | 97.33  | 98.67 | 98.67  |
| σ=18.6792  | NaN   | 90.67 | 90.67 | 94.67 | 97.33 | 94.67 | 98.67  | σ=18.6792  | NaN   | 90.67 | 94.67 | 96.00 | 97.33  | 98.67 | 98.67  |
| σ=23.8399  | NaN   | NaN   | NaN   | 93.33 | 93.33 | 96.00 | 94.67  | σ=23.8399  | NaN   | NaN   | NaN   | 96.00 | 94.67  | 98.67 | 98.67  |
| σ=30.4264  | NaN   | NaN   | 93.33 | 92.00 | 94.67 | 93.33 | 94.67  | σ=30.4264  | NaN   | NaN   | 92.00 | 94.67 | 94.67  | 97.33 | 98.67  |
| σ=38.8327  | NaN   | NaN   | NaN   | 92.00 | 94.67 | 98.67 | 94.67  | σ=38.8327  | NaN   | NaN   | NaN   | 93.33 | 96.00  | 98.67 | 98.67  |
| σ=49.5614  | NaN   | NaN   | NaN   | 90.67 | 93.33 | 94.67 | 94.67  | σ=49.5614  | NaN   | NaN   | NaN   | 93.33 | 93.33  | 98.67 | 96.00  |
| σ=63.2544  | NaN   | NaN   | NaN   | 94.67 | 90.67 | 96.00 | 97.33  | σ=63.2544  | NaN   | NaN   | NaN   | 93.33 | 90.67  | 94.67 | 97.33  |
| σ=80.7304  | NaN   | NaN   | NaN   | NaN   | 90.67 | 93.33 | 94.67  | σ=80.7304  | NaN   | NaN   | NaN   | NaN   | 93.33  | 96.00 | 96.00  |
| σ=103.0347 | NaN   | NaN   | NaN   | NaN   | 90.67 | 94.67 | 94.67  | σ=103.0347 | NaN   | NaN   | NaN   | NaN   | 89.33  | 94.67 | 96.00  |
| σ=131.5013 | NaN   | NaN   | NaN   | NaN   | NaN   | 90.67 | 94.67  | σ=131.5013 | NaN   | NaN   | NaN   | NaN   | NaN    | 93.33 | 94.67  |